



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTPP/ELECT MAINT/2023-24/LMI-EMD/543940

Date:06/10/2023(10:25:12)

Subject: LMI Generator Stator Core Faults LMI/OGN/OPS/ELEC/014

LMIIs are reviewed periodically for further improvement and incorporation of recommendation by COS.

No changes envisaged by COS . LMI has been checked and reviewed for latest updates and checklist as per latest OGN .

Put up approval of competent authority please.

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Note No:2

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Note No:4

LMI reviewed in line with OGN---LMI/OGN/OPS/ELEC/014, changes done as per OGN , forwarded for further checking and approval

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May be approved

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LMI reviewed and check list incorporated in revision, may be processed for approval, please.

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May please be accorded approval.

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**NTPC LIMITED
SIMHADRI THERMAL POWER STATION**

LOCAL MANAGEMENT INSTRUCTIONLMI/OGN/OPS/ELEC/014
ISSUE NO.: 1, REVISION-2, DATE: OCT 2023

GENERATOR STATOR CORE FAULTS

APPROVED FOR

IMPLEMENTATION BY - - - - -

HOP (SIMHADRI)

Date: _____

LMI / OGN /OPS/ ELEC / 014, ISSUE: 1, REVISION: 2, DT OCT 2023

LMI / OGN /OPS/ ELEC / 014, ISSUE: 1, REVISION: 2, DT OCT 2023

NTPC LTD.

SIMHADRI THERMAL POWER STATION

OPERATION AND MAINTENANCE DEPARTMENT

LMI(Name) :	Generator Stator Core Faults.		
LMI No.	LMI/OGN/OPS/ELEC/014		
ISSUE NO.	1	Revision : 2	Date :OCT 2023

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1.0**OBJECTIVE**

During the last few years several large generators have suffered extensive melting of the core laminations due to core faults. Experience has shown that a combination of careful erection, operation and maintenance increased awareness of all the available instrumentation that can be used on the generator, and the adoption of special procedures to deal with abnormal incidents can contribute towards failure-free operation of generators.

The foreign bodies entering the air gap may be thrown against the core bore and by re-bounding against the rotor surface, may be repeatedly brought into contact with the core. Primarily damage will be due to mechanical impact. This damage may be caused by metallic or non-metallic bodies, the former tend to be heavier and cause greater indentations. The mechanical damage will include burring over of the ends of the stamping and this will prevent a current to flow in the Iron across the burr due to increased eddy losses. Micro weld tends to occur resulting in an increase in current and temperature. Local expansion of the core can lead to other shorts across the punchings & further micro welding. Mechanical damage has a bright appearance although this can be obscured by the paint normally applied to the core. The electrical damage presents a dull smooth appearance. In extreme cases there may be evidence of burning and gross swelling of the packets, so as to close or partially close, the radial ventilation ducts.

It will assist in this field to identify generators at risk of catastrophic core faults and detail the steps to be taken to minimise the possibility of failures in Generators.

Possible Causes of Core fault during Operation**1.1.1****Leading Power Factor Operation**

Low leading power factor operations for long periods have the effect that the core end temperatures and inter laminar voltages are increased due to the increase of axial magnetic field in the stator teeth. As the load angle of the generator increases so does the magnetic field, the eddy current heating and hence the temperature. The longer the generator is left at high load angle the higher the temperature rise will be until it reaches equilibrium (usually after about one hour).

1.1.2 Sustained High Voltage (above normal) Operation

- i) High voltage, especially at leading power factors, will lead to high core end temperatures. Increased saturation of the iron core increases the axial magnetic fields. Also, the interlamination potentials are raised because the flux in the core is higher.
- ii) Where some core damage has already occurred high stator voltage operation leads to increased saturation and increased interlamination potentials, and if there is a core weakness due to, say, degraded interlamination insulation, then a core fault is more likely to occur.

1.1.3 Damage to core due to system faults

System faults that are electrically close to the generator Transformer terminals cause increased winding current, rapid changes in torque and/or electromagnetic forces; hence the contacts between the core building bars and laminations are affected. Variations in the fault forces could also distort the laminations in the tooth region, particularly where these are slack because of inadequate clamping pressure. Such distortion may lead to arc initiation between laminations by a mechanical switching process.

1.1.4 Core Faults Resulting Due to loss of synchronism

Loss of synchronism can occur with full or partial rotor current. In both cases they result in large variations of load angle which in turn give rise to rapid variations of core end heating intensity. The heat rate could lead to excessive temperature rises and enhance the risk of core faults if incipient damages already exist in the core.

2.0 SCOPE

This document covers the Generator Stator of Simhadri unit 1, 2, 3 & 4.

3.0 REFERENCE DOCUMENT

Operation information note – OIN/OPS/ELEC/014

4.0 SUPERSEDED DOCUMENT

LMI / OGN /OPS/ ELEC / 014, ISSUE: 1, REVISION: 1, DT FEB2022

Steps to be Taken during Repair/Installation/ major overhauling

5.0 5.1 Inspection

Direct visual examination of the core surface and cooling ducts should be carried out. An Introscope can be helpful in any examination of otherwise inaccessible parts of the core.

The following should be noted carefully :

- i) Debris and core fretting products (such as reddish iron oxide) or loose core plate varnish powder.
- ii) Burred, damaged, bent or loose laminations or spacers.
- iii) Burred, damaged, bent or loose laminations or spacers.
- iv) Discoloration
- v) Pitting or signs of arcing
- vi) Molten metal

5.2 Records

Records of all examinations carried out and the remedial measures undertaken should be kept with as much detail as possible. Photographs should be taken when necessary and a history of the stator should be maintained as a valuable guide for future records.

5.3 Clean Working Conditions

The areas of manufacturing assembly, site installations and repair should be maintained as clean areas.

The relevant Operation Guidance Note OGN/OPS/ELEC/002 "Guidance on Clean Conditions during Work on Generators" should be followed.

5.4 Ring Flux Tests

Interlamination insulation may be damaged during assembly or maintenance of a stator. Hot spots may also develop during operation due to ingress of foreign bodies into the bore, or the deterioration of interlamination core stamping insulation. If allowed to persist, hot spots can damage the insulation on the adjacent conductor bars. They should therefore be detected and if possible repaired during routine overhauls. The following techniques are used for determining the detection of core faults.

5.4.1 Thermal Ring Flux Test :

Stator cores can be tested for damage using a thermal ring flux test in which the core is excited to give about 80% of the rated flux. Any hot spots on the teeth tops are readily detected with thermo vision camera scanning the bore, but the test is unlikely to detect deep seated faults.

5.4.2 EL-CID Test (Electromagnetic Core imperfection Detector Test):

This employs the modern technique of detecting the core faults. In this test the core is excited by much lower power, which gives about 4% of the rated flux in the core. The voltage gradient axially along the core induces currents to flow

through any damaged areas but because the current is small therefore the heating is insignificant. The test relies on the electromagnetic detection of these fault currents following axially through the damaged region "This test enables to detect deep seated core faults in the generator core."

The above mentioned techniques for detecting the core faults, should be applied in the following circumstances.

- i) When the generator has been built in the manufacturers works.
- ii) When visible damage is present and repair work is carried out.
- iii) When there is indirect evidence of damage, such as hot spots.
- iv) When it has been decided for some other reasons to check on the condition of the core, for example a periodic health checking of core as a part of preventive maintenance requirement and capital overhauls.

Operation Information Note "Ring flux testing of Generator stator cores" describing both Ring flux & ELCID Test should be consulted

6- Steps to be Taken during Operation to Prevent Core Failure

6.1 Temperature Measurements

During commissioning and loading stages, attempts should be made to establish the interdependence of the temperature rises with electrical quantities and also which thermocouples give the highest temperatures. (They are generally in the core end region). They should be checked frequently, preferably by continuous monitoring to ensure that the temperature limits are not exceeded. Any abnormality should be noted and investigated, with reference to manufacturers works test reports.

Expected temperature rise limits recommended by the manufacturers should be adhered to during the operation. Where possible, alarms should be set to signify temperature limits.

6.2 Excitation System and Stability Limit

- i) There should be enough margin of stability limit in the generator operating diagram when operating in manual excitation control, so that the likelihood of pole-slip will be avoided when under manual excitation control, the load is increased either by manual interventions or due to system fluctuations.
- ii) When the generator is operating with manually controlled excitation, extreme care should be taken to avoid sudden changes of load, so that there is always time to make appropriate changes in excitation.
- iii) Every generator should have its own Vector Diagram Displayed on the Control Desk with the stability limits, both for AVR and manual operation, clearly marked. The operator should be advised always to operate the generator within these limits.
- iv) During AVR changeover it is important that a switch to manual is only made when the balance indication is zero and has been observed in this condition for at least five seconds.
- v) It should be ensured that the manual excitation setting should not be lower than that of the automatic (AVR) setting.
- vi) As soon as pole slipping conditions is observed caution is required to watch whether increasing excitation will correct the condition and if such correction does not take place then tripping of the unit is the only alternative to avoid damage.

vii) Pole-slipping protection, if fitted, should be set to operate before the generator slips one pole pair. If pole-slipping protection is not installed, the settings of the loss of synchronisation protection should be so chosen as to respond most effectively to pole-slipping.

6.3 Events Recording

It is important that regular records of stator current, voltage, MW, MVAR and load angle should be made at appropriate intervals. The operation of all alarms, indicators and protection flags should be recorded. System faults and disturbances together with incidents affecting generator stability and loss of stability should also be recorded

6.4 Core Monitors

Core Conditions/Gas Condition Monitors CCM/GCM have been developed by some specialist firms and are Commercially available. The necessity of providing, such core monitors as a condition monitoring device, needs to be decided from case to case basis. However, this device should have provision for the following.

- i) Particle Detectors (PD), to monitor the condition of the core as well as the state of winding insulation; they should include automatic filter samplers.
- ii) The facility to analyze the ambient hydrogen gas inside the casing.
- iii) Regular hydrogen gas analysis carried out to check the condition of the core as well as the quality of the hydrogen gas being used.

6.5 Core Monitor Alarm

It is recommended strictly follow the instructions provided by the supplier of the core monitor. However when a core monitor alarm is received the following action should be taken:

- i) Verify that the instrument is functioning correctly using the built-in check facilities.
- ii) Take hydrogen gas samples for analysis.
- iii) Change the loading conditions.

If after reducing the load by 20 percent the core monitor alarm still exists, then contact the Deputy General Manager (Operation) for advice on further load reduction or Tripping of the Unit. In any case, the load must be progressively reduced until the alarming condition clears and ultimately shut down if alarm persists, because irreparable damage can take place if faulty core is kept energised. Action to restart the unit should be based on analysis and proper validation of the alarm.

- v) Arrange to get sample analysis as recommended by supplier, to establish validity of damaging condition.

7. Action to be Taken When a Generator has pole-slipped or Subjected to Short-Circuit.

- i) Examine the records to verify that pole-slip took place.
- ii) At convenient time, such as during a planned outage, the generator should be given a ring flux test to find out if there are any hotspots.
- iii) Until the generator can be checked for hotspots, its operation should be continued with the assumption that there is an incipient core fault.
- iv) Unless there is evidence of hotspots, such as a core monitor alarm, the only operating restrictions are the avoidance of leading power factors and

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high terminal voltages. The load should be kept as steady as possible and any change should be carried out gradually to minimise the torque fluctuations on the stator core. When on manual excitation, ensure the excitation is increased before the application of the load.

v) Special vigilance must be kept for further abnormal conditions such as pole-slips and loss of Synchronisation.

vi) Other tests should also be carried out to check the integrity of the rotor winding following a pole-slipping incident.

8-FREQUENCY OF CHECKING AND MEASUREMENTS:

During Generator major overhauling

9- RESPONSIBILITY CENTERS FOR IMPLEMENTATION:

Instruction	Responsibility
Responsibility for implementation	DGM (EM-Generator group)
Temperature monitoring and Event recording	Operation shift

10- EXCEPTION REPORTING & DISPENSATION OF DEVIATIONS:

Any deviation shall be reported to HOD (EM).

11- RECORD KEEPING:

Records will be kept with Generator group of Electrical Maintenance.

12- REVIEW:

Document will be reviewed after two years by HOP (Simhadri).

13-RECORDS OF REVISION:

LMI modified as per OIN/OPS/ELEC/014

CHECKLIST DURING OVERHAUL**UNIT:****DATE:**

S.NO	DESCRIPTION	OBSERVATION/LIMITS	REMARKS
1	Visual examination for any debris and core fretting products (such as reddish iron oxide) or loose core plate varnish powder		
2	Any burred, damaged, bent or loose laminations or spacers observed		
3	Visual inspection for any Discoloration		
4	Visual inspection for Pitting or signs of arcing		
5	Visual inspection for Molten metal		
6	Any abnormality in ELCID test	Leakage current should be less than 100 milli amps	